FMv2 ODE Steps=10 Analysis

Three-way comparison — A: Diffusion S6 · E: FMv2 ODE=1 (default) · F: FMv2 ODE=10

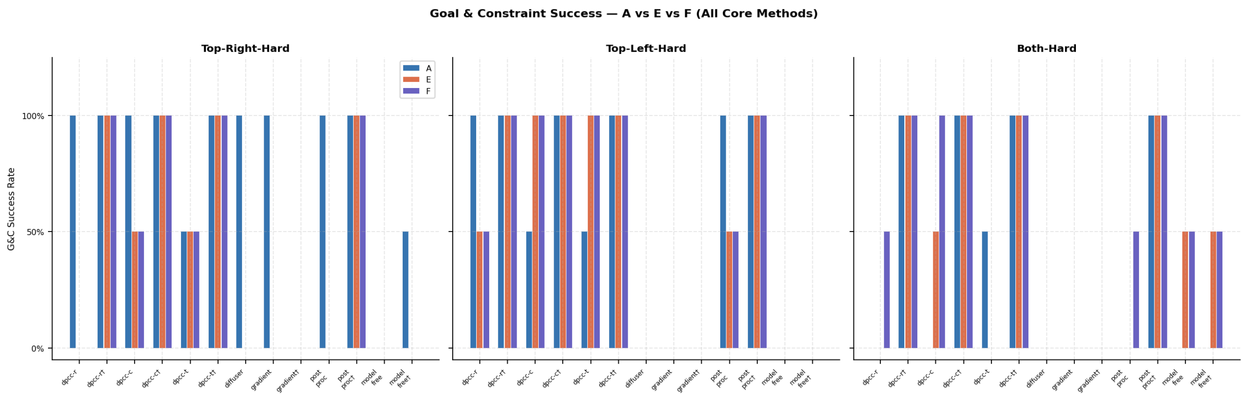
A = Diffusion S6 · baseline reference	**E = FMv2 ODE=1 · new arch, default eval**	**F = FMv2 ODE=10 · new arch, ODE steps=10**

F uses the same FMv2 architecture and action_weight=1 as E, but with **ODE steps=10** at evaluation time (E uses the default ODE steps=1). Same model checkpoint (step 90000), same Seed 6. The core question: does increasing ODE integration steps improve trajectory quality or constraint satisfaction, and at what computational cost?

Section 1 — Visual Analysis: A vs E vs F

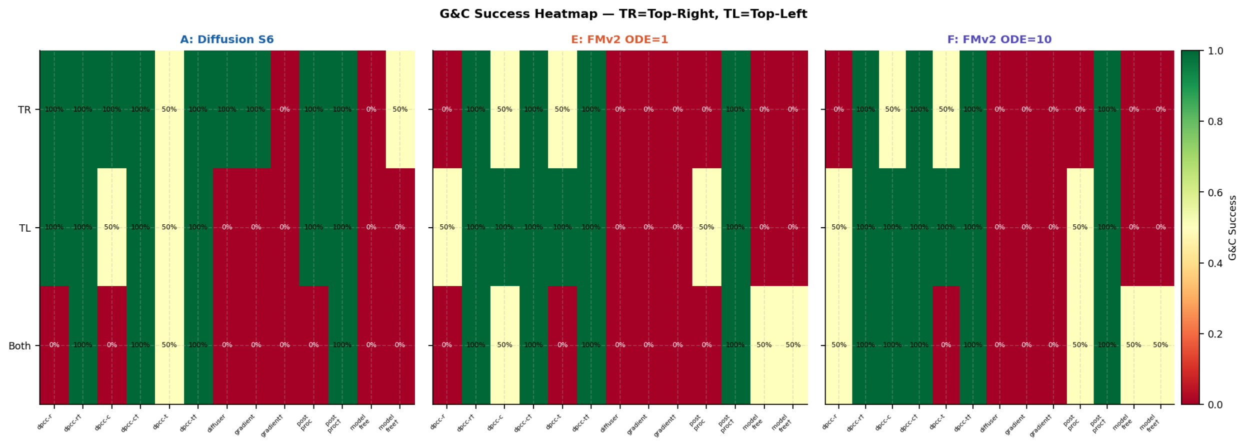
1.1 Goal & Constraint Success — All Core Methods

Primary metric. Blue=A (Diffusion), coral=E (FMv2 ODE=1), purple=F (FMv2 ODE=10). †=tightened.



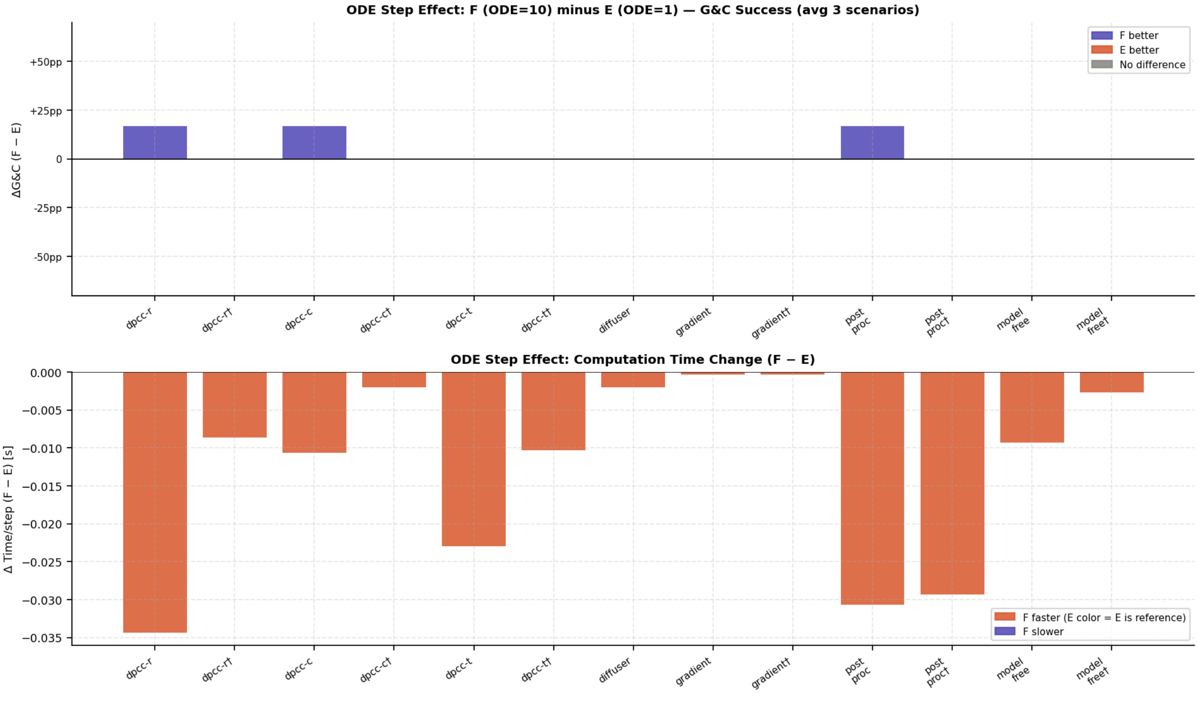
1.2 G&C; Heatmap — All 3 Configurations

Green=100%, yellow=50%, red=0%. TR=Top-Right, TL=Top-Left.



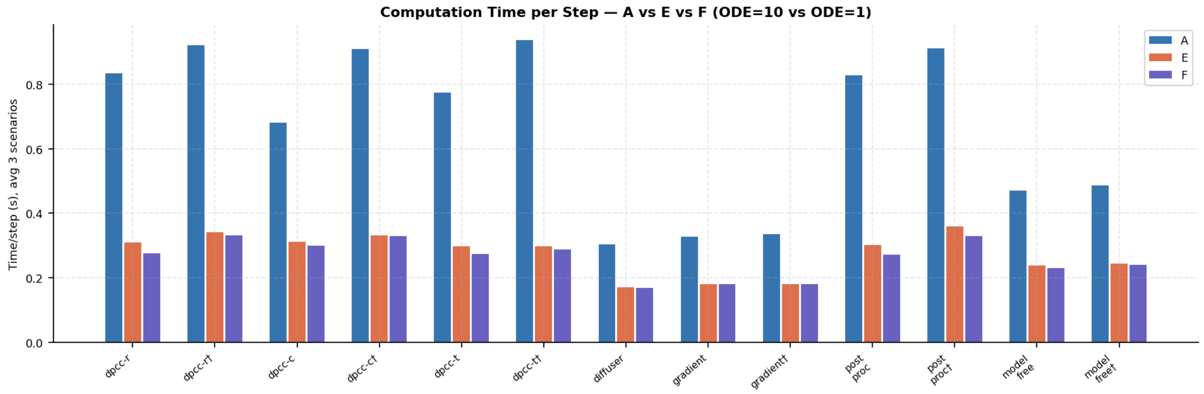
1.3 ODE Step Effect: F minus E — G&C; Delta and Time Delta

Top panel: G&C; change from increasing ODE steps 1→10 (avg 3 scenarios). Purple=F improves, coral=E was better, gray=no change. Bottom panel: computation time change. Most differences are near zero — the ODE step count has minimal effect on both quality and speed for this architecture.



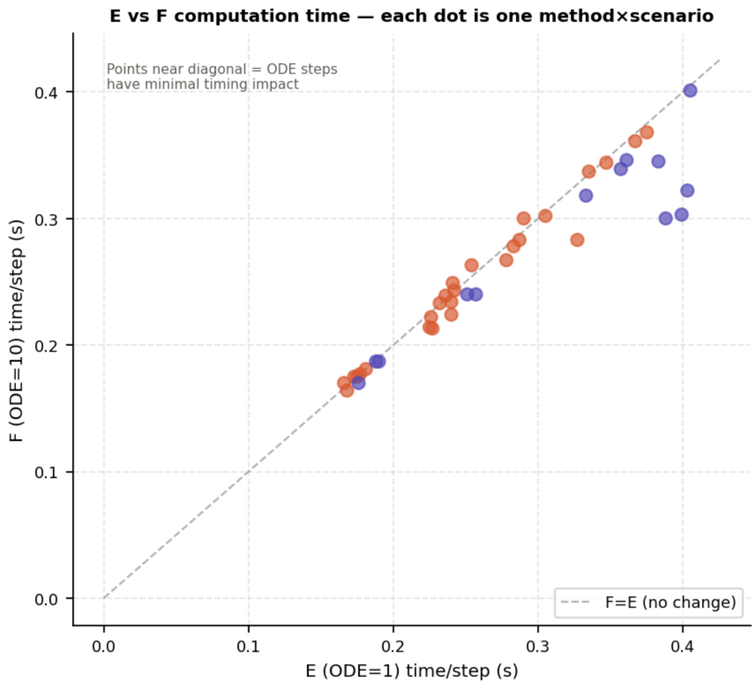
1.4 Computation Time per Step — A vs E vs F

Absolute time values averaged over 3 scenarios. E and F are nearly identical in speed.



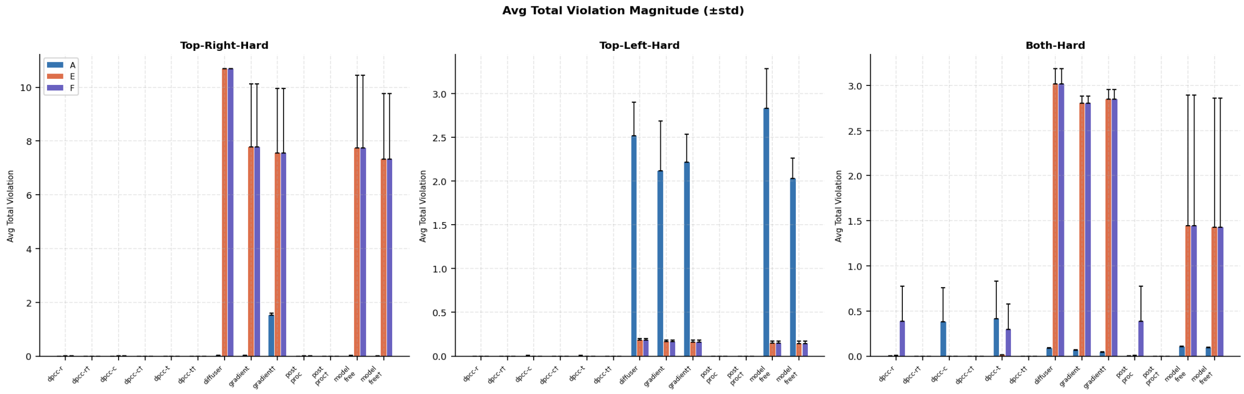
1.5 E vs F Time Scatter — Each Point is One Method × Scenario

Points near the diagonal confirm E and F have near-identical computation times. Only DPCC methods with heavy projection (dt=0.25) deviate noticeably.

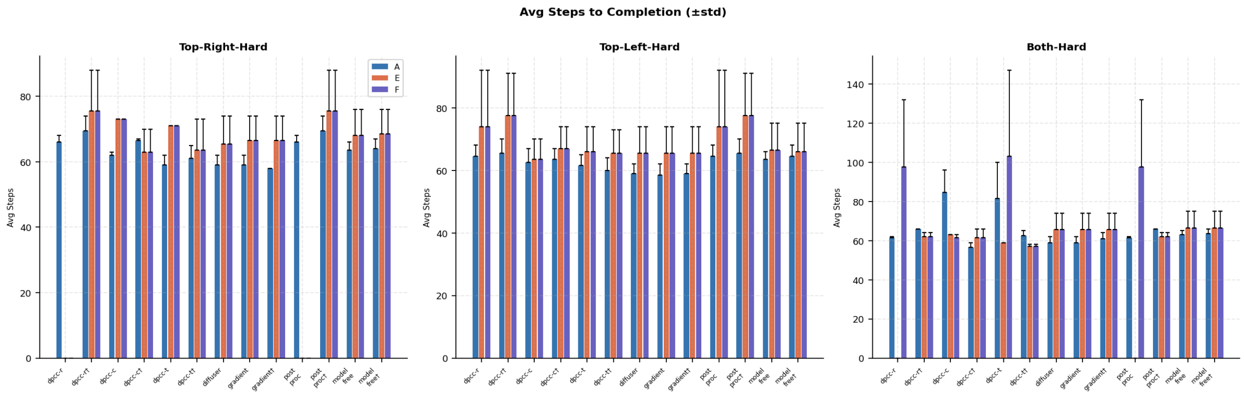


1.6 Avg Constraint Violations (±std)

1.7 Avg Total Violation Magnitude (±std)

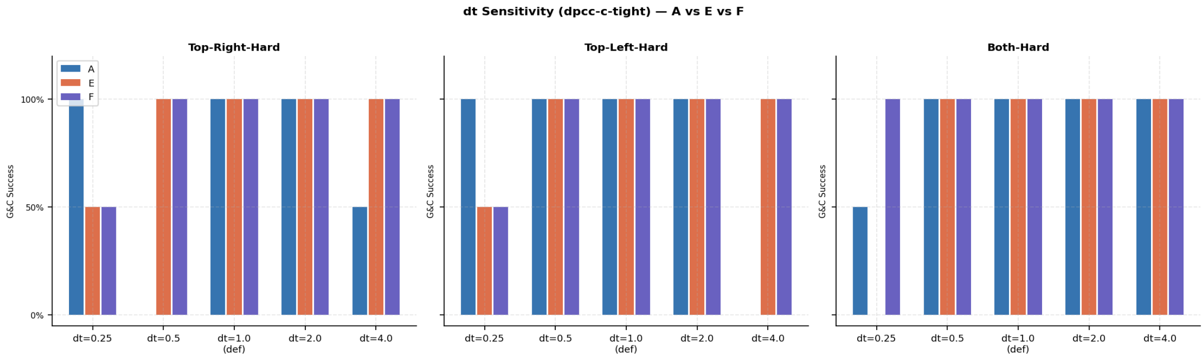


1.8 Avg Steps to Completion (±std)

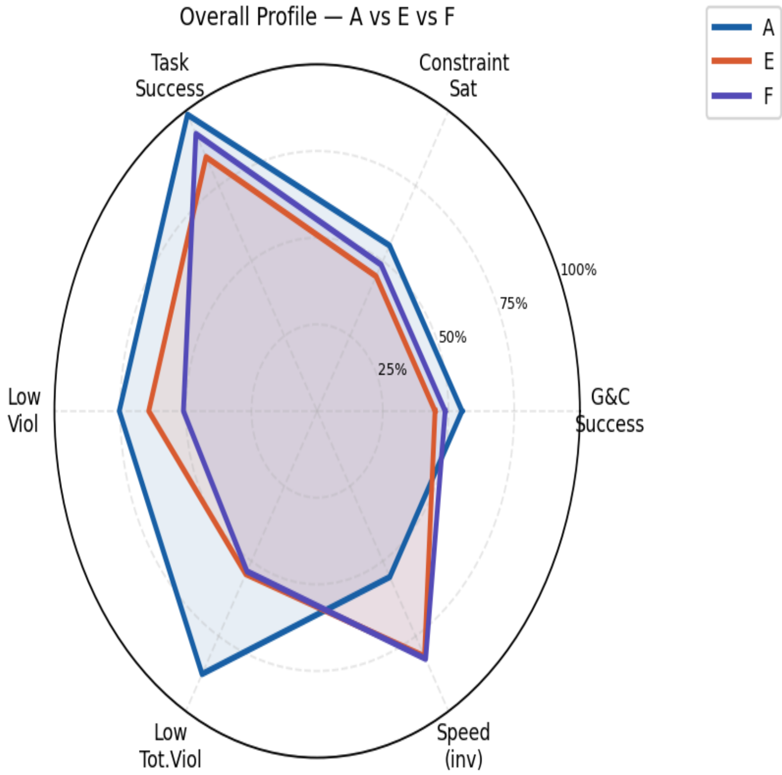


1.9 dt Sensitivity — dpcc-c-tightened: A vs E vs F

G&C; success across dt ∈ {0.25, 0.5, 1.0, 2.0, 4.0}. F fixes E's dt=0.25 failure on both-hard (0%→100%).



1.10 Overall Model Profile Radar



Section 2 — Complete Raw Data Tables

All metrics for F (FMv2 ODE=10), E, and A. **SR**=Success Rate, **CS**=Constraints Satisfied, **G&C**=Goal & Constraints, **Steps±std**, **Viol#±std**, **Viol∑±std**, **Time**(s/step), **TE**=Tracking Error.

2.1 Top-Right-Hard

F: FMv2 ODE=10 — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	0%	0%	0%	0.0±0.0	1.5±1.5	0.008±0.008	0.222	–
dpcc-r-tight	100%	100%	100%	75.5±12.5	0.0±0.0	0.000±0.000	0.283	–
dpcc-c	50%	50%	50%	73.0±0.0	1.0±1.0	0.001±0.001	0.224	–
dpcc-c-tight	100%	100%	100%	63.0±7.0	0.0±0.0	0.000±0.000	0.243	–
dpcc-t	50%	50%	50%	71.0±0.0	0.0±0.0	0.000±0.000	0.239	–
dpcc-t-tight	100%	100%	100%	63.5±9.5	0.0±0.0	0.000±0.000	0.267	–
diffuser	100%	0%	0%	65.5±8.5	33.0±1.0	10.682±0.010	0.164	0.029
gradient	100%	0%	0%	66.5±7.5	34.0±2.0	7.776±2.344	0.175	–
gradient-tight	100%	0%	0%	66.5±7.5	34.0±2.0	7.560±2.399	0.175	–
post_proc	0%	0%	0%	0.0±0.0	1.5±1.5	0.008±0.008	0.213	–
post_proc-tight	100%	100%	100%	75.5±12.5	0.0±0.0	0.000±0.000	0.283	–
model_free	100%	0%	0%	68.0±8.0	35.5±1.5	7.743±2.693	0.214	–
model_free-tight	100%	0%	0%	68.5±7.5	36.0±2.0	7.332±2.428	0.233	–
dpcc-c-tight-dt0.25	100%	50%	50%	90.5±12.5	1.0±1.0	0.003±0.003	0.452	–
dpcc-c-tight-dt0.5	100%	100%	100%	72.0±11.0	0.0±0.0	0.000±0.000	0.275	–
dpcc-c-tight-dt2.0	100%	100%	100%	72.5±0.5	0.0±0.0	0.000±0.000	0.248	–
dpcc-c-tight-dt4.0	100%	100%	100%	142.0±4.0	0.0±0.0	0.000±0.000	0.260	–

E: FMv2 ODE=1 — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	0%	0%	0%	0.0±0.0	1.5±1.5	0.008±0.008	0.226	–
dpcc-r-tight	100%	100%	100%	75.5±12.5	0.0±0.0	0.000±0.000	0.287	–
dpcc-c	50%	50%	50%	73.0±0.0	1.0±1.0	0.001±0.001	0.240	–
dpcc-c-tight	100%	100%	100%	63.0±7.0	0.0±0.0	0.000±0.000	0.242	–
dpcc-t	50%	50%	50%	71.0±0.0	0.0±0.0	0.000±0.000	0.236	–
dpcc-t-tight	100%	100%	100%	63.5±9.5	0.0±0.0	0.000±0.000	0.278	–
diffuser	100%	0%	0%	65.5±8.5	33.0±1.0	10.682±0.010	0.168	0.029
gradient	100%	0%	0%	66.5±7.5	34.0±2.0	7.776±2.344	0.173	–
gradient-tight	100%	0%	0%	66.5±7.5	34.0±2.0	7.560±2.399	0.175	–
post_proc	0%	0%	0%	0.0±0.0	1.5±1.5	0.008±0.008	0.227	–
post_proc-tight	100%	100%	100%	75.5±12.5	0.0±0.0	0.000±0.000	0.327	–
model_free	100%	0%	0%	68.0±8.0	35.5±1.5	7.743±2.693	0.225	–
model_free-tight	100%	0%	0%	68.5±7.5	36.0±2.0	7.332±2.428	0.232	–
dpcc-c-tight-dt0.25	100%	50%	50%	90.5±12.5	1.0±1.0	0.003±0.003	0.453	–

dpcc-c-tight-dt0.5	100%	100%	100%	72.0±11.0	0.0±0.0	0.000±0.000	0.274	–
dpcc-c-tight-dt2.0	100%	100%	100%	72.5±0.5	0.0±0.0	0.000±0.000	0.247	–
dpcc-c-tight-dt4.0	100%	100%	100%	142.0±4.0	0.0±0.0	0.000±0.000	0.268	–

A: Diffusion — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	100%	100%	66.0±2.0	0.0±0.0	0.000±0.000	0.698	–
dpcc-r-tight	100%	100%	100%	69.5±4.5	0.0±0.0	0.000±0.000	0.817	–
dpcc-c	100%	100%	100%	62.0±1.0	0.0±0.0	0.000±0.000	0.744	–
dpcc-c-tight	100%	100%	100%	66.5±0.5	0.0±0.0	0.000±0.000	0.911	–
dpcc-t	100%	50%	50%	59.0±3.0	0.5±0.5	0.000±0.000	0.799	–
dpcc-t-tight	100%	100%	100%	61.0±4.0	0.0±0.0	0.000±0.000	0.871	–
diffuser	100%	100%	100%	59.0±3.0	0.0±0.0	0.021±0.002	0.298	0.025
gradient	100%	100%	100%	59.0±3.0	0.0±0.0	0.021±0.002	0.316	–
gradient-tight	50%	0%	0%	58.0±0.0	15.0±0.0	1.521±0.082	0.327	–
post_proc	100%	100%	100%	66.0±2.0	0.0±0.0	0.000±0.000	0.670	–
post_proc-tight	100%	100%	100%	69.5±4.5	0.0±0.0	0.000±0.000	0.814	–
model_free	100%	0%	0%	63.5±2.5	3.0±1.0	0.012±0.011	0.490	–
model_free-tight	100%	50%	50%	64.0±3.0	2.0±2.0	0.009±0.009	0.508	–
dpcc-c-tight-dt0.25	100%	100%	100%	81.0±6.0	0.0±0.0	0.002±0.000	2.225	–
dpcc-c-tight-dt0.5	100%	0%	0%	75.5±2.5	1.0±0.0	0.003±0.000	1.668	–
dpcc-c-tight-dt2.0	100%	100%	100%	66.5±0.5	0.0±0.0	0.000±0.000	0.689	–
dpcc-c-tight-dt4.0	50%	50%	50%	72.0±0.0	0.0±0.0	0.000±0.000	0.535	–

2.2 Top-Left-Hard

F: FMv2 ODE=10 — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	50%	50%	74.0±18.0	0.5±0.5	0.000±0.000	0.302	–
dpcc-r-tight	100%	100%	100%	77.5±13.5	0.0±0.0	0.000±0.000	0.368	–
dpcc-c	100%	100%	100%	63.5±6.5	0.0±0.0	0.000±0.000	0.337	–
dpcc-c-tight	100%	100%	100%	67.0±7.0	0.0±0.0	0.000±0.000	0.344	–
dpcc-t	100%	100%	100%	66.0±8.0	0.0±0.0	0.000±0.000	0.263	–
dpcc-t-tight	100%	100%	100%	65.5±7.5	0.0±0.0	0.000±0.000	0.278	–
diffuser	100%	0%	0%	65.5±8.5	10.0±1.0	0.181±0.022	0.170	0.029
gradient	100%	0%	0%	65.5±8.5	10.0±1.0	0.165±0.018	0.177	–
gradient-tight	100%	0%	0%	65.5±8.5	10.0±1.0	0.162±0.021	0.181	–
post_proc	100%	50%	50%	74.0±18.0	0.5±0.5	0.000±0.000	0.300	–
post_proc-tight	100%	100%	100%	77.5±13.5	0.0±0.0	0.000±0.000	0.361	–
model_free	100%	0%	0%	66.5±8.5	10.0±1.0	0.146±0.026	0.234	–
model_free-tight	100%	0%	0%	66.0±9.0	10.0±1.0	0.143±0.028	0.249	–
dpcc-c-tight-dt0.25	100%	50%	50%	137.5±37.5	51.5±51.5	0.840±0.840	2.577	–

dpcc-c-tight-dt0.5	100%	100%	100%	76.0±9.0	0.0±0.0	0.001±0.000	0.523	–
dpcc-c-tight-dt2.0	100%	100%	100%	69.5±3.5	0.0±0.0	0.000±0.000	0.267	–
dpcc-c-tight-dt4.0	100%	100%	100%	142.5±8.5	0.0±0.0	0.000±0.000	0.248	–

E: FMv2 ODE=1 — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	50%	50%	74.0±18.0	0.5±0.5	0.000±0.000	0.305	–
dpcc-r-tight	100%	100%	100%	77.5±13.5	0.0±0.0	0.000±0.000	0.375	–
dpcc-c	100%	100%	100%	63.5±6.5	0.0±0.0	0.000±0.000	0.335	–
dpcc-c-tight	100%	100%	100%	67.0±7.0	0.0±0.0	0.000±0.000	0.347	–
dpcc-t	100%	100%	100%	66.0±8.0	0.0±0.0	0.000±0.000	0.254	–
dpcc-t-tight	100%	100%	100%	65.5±7.5	0.0±0.0	0.000±0.000	0.283	–
diffuser	100%	0%	0%	65.5±8.5	10.0±1.0	0.181±0.022	0.166	0.029
gradient	100%	0%	0%	65.5±8.5	10.0±1.0	0.165±0.018	0.177	–
gradient-tight	100%	0%	0%	65.5±8.5	10.0±1.0	0.162±0.021	0.181	–
post_proc	100%	50%	50%	74.0±18.0	0.5±0.5	0.000±0.000	0.290	–
post_proc-tight	100%	100%	100%	77.5±13.5	0.0±0.0	0.000±0.000	0.367	–
model_free	100%	0%	0%	66.5±8.5	10.0±1.0	0.146±0.026	0.240	–
model_free-tight	100%	0%	0%	66.0±9.0	10.0±1.0	0.143±0.028	0.241	–
dpcc-c-tight-dt0.25	100%	50%	50%	137.5±37.5	51.5±51.5	0.840±0.840	2.627	–
dpcc-c-tight-dt0.5	100%	100%	100%	76.0±9.0	0.0±0.0	0.001±0.000	0.517	–
dpcc-c-tight-dt2.0	100%	100%	100%	69.5±3.5	0.0±0.0	0.000±0.000	0.259	–
dpcc-c-tight-dt4.0	100%	100%	100%	142.5±8.5	0.0±0.0	0.000±0.000	0.246	–

A: Diffusion — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	100%	100%	64.5±3.5	0.0±0.0	0.000±0.000	0.784	–
dpcc-r-tight	100%	100%	100%	65.5±4.5	0.0±0.0	0.000±0.000	0.998	–
dpcc-c	100%	50%	50%	62.5±4.5	1.0±1.0	0.003±0.003	0.592	–
dpcc-c-tight	100%	100%	100%	63.5±3.5	0.0±0.0	0.000±0.000	0.822	–
dpcc-t	100%	50%	50%	61.5±3.5	1.5±1.5	0.004±0.004	0.829	–
dpcc-t-tight	100%	100%	100%	60.0±4.0	0.0±0.0	0.000±0.000	0.976	–
diffuser	100%	0%	0%	59.0±3.0	20.0±2.0	2.515±0.387	0.306	0.025
gradient	100%	0%	0%	58.5±3.5	17.5±3.5	2.116±0.571	0.334	–
gradient-tight	100%	0%	0%	59.0±3.0	19.5±1.5	2.218±0.316	0.348	–
post_proc	100%	100%	100%	64.5±3.5	0.0±0.0	0.000±0.000	0.782	–
post_proc-tight	100%	100%	100%	65.5±4.5	0.0±0.0	0.000±0.000	0.996	–
model_free	100%	0%	0%	63.5±2.5	19.0±0.0	2.829±0.455	0.458	–
model_free-tight	100%	0%	0%	64.5±3.5	15.0±2.0	2.029±0.233	0.470	–
dpcc-c-tight-dt0.25	100%	100%	100%	70.0±3.0	0.0±0.0	0.000±0.000	1.259	–
dpcc-c-tight-dt0.5	100%	100%	100%	63.5±0.5	0.0±0.0	0.000±0.000	0.937	–

dpcc-c-tight-dt2.0	100%	100%	100%	64.5±0.5	0.0±0.0	0.000±0.000	0.679	–
dpcc-c-tight-dt4.0	100%	0%	0%	107.0±2.0	16.5±0.5	3.711±0.053	0.810	–

2.3 Both-Hard

F: FMv2 ODE=10 — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	50%	50%	97.5±34.5	36.5±36.5	0.388±0.388	0.303	–
dpcc-r-tight	100%	100%	100%	62.0±2.0	0.0±0.0	0.000±0.000	0.346	–
dpcc-c	100%	100%	100%	61.5±1.5	0.0±0.0	0.000±0.000	0.339	–
dpcc-c-tight	100%	100%	100%	61.5±4.5	0.0±0.0	0.000±0.000	0.401	–
dpcc-t	100%	0%	0%	103.0±44.0	35.5±32.5	0.294±0.283	0.322	–
dpcc-t-tight	100%	100%	100%	57.0±1.0	0.0±0.0	0.000±0.000	0.318	–
diffuser	100%	0%	0%	65.5±8.5	11.5±0.5	3.013±0.172	0.170	0.029
gradient	100%	0%	0%	65.5±8.5	11.5±0.5	2.804±0.074	0.187	–
gradient-tight	100%	0%	0%	65.5±8.5	12.0±1.0	2.844±0.109	0.187	–
post_proc	100%	50%	50%	97.5±34.5	36.5±36.5	0.388±0.388	0.300	–
post_proc-tight	100%	100%	100%	62.0±2.0	0.0±0.0	0.000±0.000	0.345	–
model_free	100%	50%	50%	66.5±8.5	6.0±6.0	1.445±1.445	0.240	–
model_free-tight	100%	50%	50%	66.5±8.5	6.0±6.0	1.430±1.430	0.240	–
dpcc-c-tight-dt0.25	100%	100%	100%	71.5±3.5	0.0±0.0	0.000±0.000	0.572	–
dpcc-c-tight-dt0.5	100%	100%	100%	60.5±5.5	0.0±0.0	0.000±0.000	0.556	–
dpcc-c-tight-dt2.0	100%	100%	100%	72.0±14.0	0.0±0.0	0.000±0.000	0.319	–
dpcc-c-tight-dt4.0	100%	100%	100%	96.0±20.0	0.0±0.0	0.000±0.000	0.315	–

E: FMv2 ODE=1 — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	0%	0%	0%	0.0±0.0	1.0±1.0	0.005±0.005	0.399	–
dpcc-r-tight	100%	100%	100%	62.0±2.0	0.0±0.0	0.000±0.000	0.361	–
dpcc-c	50%	50%	50%	63.0±0.0	0.0±0.0	0.000±0.000	0.357	–
dpcc-c-tight	100%	100%	100%	61.5±4.5	0.0±0.0	0.000±0.000	0.405	–
dpcc-t	50%	0%	0%	59.0±0.0	3.5±0.5	0.013±0.002	0.403	–
dpcc-t-tight	100%	100%	100%	57.0±1.0	0.0±0.0	0.000±0.000	0.333	–
diffuser	100%	0%	0%	65.5±8.5	11.5±0.5	3.013±0.172	0.176	0.029
gradient	100%	0%	0%	65.5±8.5	11.5±0.5	2.804±0.074	0.190	–
gradient-tight	100%	0%	0%	65.5±8.5	12.0±1.0	2.844±0.109	0.188	–
post_proc	0%	0%	0%	0.0±0.0	1.0±1.0	0.005±0.005	0.388	–
post_proc-tight	100%	100%	100%	62.0±2.0	0.0±0.0	0.000±0.000	0.383	–
model_free	100%	50%	50%	66.5±8.5	6.0±6.0	1.445±1.445	0.251	–
model_free-tight	100%	50%	50%	66.5±8.5	6.0±6.0	1.430±1.430	0.257	–
dpcc-c-tight-dt0.25	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.693	–
dpcc-c-tight-dt0.5	100%	100%	100%	60.5±5.5	0.0±0.0	0.000±0.000	0.585	–

dpcc-c-tight-dt2.0	100%	100%	100%	72.0±14.0	0.0±0.0	0.000±0.000	0.336	–
dpcc-c-tight-dt4.0	100%	100%	100%	96.0±20.0	0.0±0.0	0.000±0.000	0.324	–

A: Diffusion — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	0%	0%	61.5±0.5	2.5±0.5	0.002±0.001	1.019	–
dpcc-r-tight	100%	100%	100%	66.0±0.0	0.0±0.0	0.000±0.000	0.948	–
dpcc-c	100%	0%	0%	84.5±11.5	19.0±17.0	0.381±0.377	0.704	–
dpcc-c-tight	100%	100%	100%	56.5±2.5	0.0±0.0	0.000±0.000	0.997	–
dpcc-t	100%	50%	50%	81.5±18.5	17.0±17.0	0.416±0.416	0.697	–
dpcc-t-tight	100%	100%	100%	62.5±2.5	0.0±0.0	0.000±0.000	0.964	–
diffuser	100%	0%	0%	59.0±3.0	7.5±0.5	0.090±0.001	0.307	0.025
gradient	100%	0%	0%	59.0±3.0	7.0±1.0	0.064±0.007	0.331	–
gradient-tight	100%	0%	0%	61.0±3.0	6.0±1.0	0.044±0.004	0.332	–
post_proc	100%	0%	0%	61.5±0.5	2.5±0.5	0.002±0.001	1.029	–
post_proc-tight	100%	100%	100%	66.0±0.0	0.0±0.0	0.000±0.000	0.924	–
model_free	100%	0%	0%	63.0±2.0	8.5±0.5	0.103±0.008	0.465	–
model_free-tight	100%	0%	0%	63.5±2.5	8.5±0.5	0.091±0.010	0.481	–
dpcc-c-tight-dt0.25	100%	50%	50%	75.5±0.5	0.5±0.5	0.000±0.000	1.835	–
dpcc-c-tight-dt0.5	100%	100%	100%	68.5±4.5	0.0±0.0	0.000±0.000	1.185	–
dpcc-c-tight-dt2.0	100%	100%	100%	64.5±2.5	0.0±0.0	0.000±0.000	0.737	–
dpcc-c-tight-dt4.0	100%	100%	100%	85.0±2.0	0.0±0.0	0.000±0.000	0.691	–

Section 3 — Three-Way Head-to-Head Comparison

Best G&C; highlighted green. Best (lowest) violation count highlighted teal. ΔEF = F G&C; minus E G&C; (shows ODE step impact per method).

Top-Right-Hard

Method	A G&C;	E G&C;	F G&C;	Best	ΔEF	A Viol	E Viol	F Viol	E Time	F Time
dpcc-r	100%	0%	0%	A	=	0.0	1.5	1.5	0.226	0.222
dpcc-r-tight	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.287	0.283
dpcc-c	100%	50%	50%	A	=	0.0	1.0	1.0	0.240	0.224
dpcc-c-tight	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.242	0.243
dpcc-t	50%	50%	50%	A/E/F	=	0.5	0.0	0.0	0.236	0.239
dpcc-t-tight	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.278	0.267
diffuser	100%	0%	0%	A	=	0.0	33.0	33.0	0.168	0.164
gradient	100%	0%	0%	A	=	0.0	34.0	34.0	0.173	0.175
gradient-tight	0%	0%	0%	A/E/F	=	15.0	34.0	34.0	0.175	0.175
post_proc	100%	0%	0%	A	=	0.0	1.5	1.5	0.227	0.213
post_proc-tight	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.327	0.283
model_free	0%	0%	0%	A/E/F	=	3.0	35.5	35.5	0.225	0.214
model_free-tight	50%	0%	0%	A	=	2.0	36.0	36.0	0.232	0.233
dpcc-c-tight-dt0.25	100%	50%	50%	A	=	0.0	1.0	1.0	0.453	0.452
dpcc-c-tight-dt0.5	0%	100%	100%	E/F	=	1.0	0.0	0.0	0.274	0.275
dpcc-c-tight-dt2.0	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.247	0.248
dpcc-c-tight-dt4.0	50%	100%	100%	E/F	=	0.0	0.0	0.0	0.268	0.260

Top-Left-Hard

Method	A G&C;	E G&C;	F G&C;	Best	ΔEF	A Viol	E Viol	F Viol	E Time	F Time
dpcc-r	100%	50%	50%	A	=	0.0	0.5	0.5	0.305	0.302
dpcc-r-tight	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.375	0.368
dpcc-c	50%	100%	100%	E/F	=	1.0	0.0	0.0	0.335	0.337
dpcc-c-tight	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.347	0.344
dpcc-t	50%	100%	100%	E/F	=	1.5	0.0	0.0	0.254	0.263
dpcc-t-tight	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.283	0.278
diffuser	0%	0%	0%	A/E/F	=	20.0	10.0	10.0	0.166	0.170
gradient	0%	0%	0%	A/E/F	=	17.5	10.0	10.0	0.177	0.177
gradient-tight	0%	0%	0%	A/E/F	=	19.5	10.0	10.0	0.181	0.181
post_proc	100%	50%	50%	A	=	0.0	0.5	0.5	0.290	0.300
post_proc-tight	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.367	0.361
model_free	0%	0%	0%	A/E/F	=	19.0	10.0	10.0	0.240	0.234
model_free-tight	0%	0%	0%	A/E/F	=	15.0	10.0	10.0	0.241	0.249

dpcc-c-tight-dt0.25	100%	50%	50%	A	=	0.0	51.5	51.5	2.627	2.577
dpcc-c-tight-dt0.5	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.517	0.523
dpcc-c-tight-dt2.0	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.259	0.267
dpcc-c-tight-dt4.0	0%	100%	100%	E/F	=	16.5	0.0	0.0	0.246	0.248

Both-Hard

Method	A G&C;	E G&C;	F G&C;	Best	∆EF	A Viol	E Viol	F Viol	E Time	F Time
dpcc-r	0%	0%	50%	F	+50pp	2.5	1.0	36.5	0.399	0.303
dpcc-r-tight	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.361	0.346
dpcc-c	0%	50%	100%	F	+50pp	19.0	0.0	0.0	0.357	0.339
dpcc-c-tight	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.405	0.401
dpcc-t	50%	0%	0%	A	=	17.0	3.5	35.5	0.403	0.322
dpcc-t-tight	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.333	0.318
diffuser	0%	0%	0%	A/E/F	=	7.5	11.5	11.5	0.176	0.170
gradient	0%	0%	0%	A/E/F	=	7.0	11.5	11.5	0.190	0.187
gradient-tight	0%	0%	0%	A/E/F	=	6.0	12.0	12.0	0.188	0.187
post_proc	0%	0%	50%	F	+50pp	2.5	1.0	36.5	0.388	0.300
post_proc-tight	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.383	0.345
model_free	0%	50%	50%	E/F	=	8.5	6.0	6.0	0.251	0.240
model_free-tight	0%	50%	50%	E/F	=	8.5	6.0	6.0	0.257	0.240
dpcc-c-tight-dt0.25	50%	0%	100%	F	+100pp	0.5	0.0	0.0	0.693	0.572
dpcc-c-tight-dt0.5	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.585	0.556
dpcc-c-tight-dt2.0	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.336	0.319
dpcc-c-tight-dt4.0	100%	100%	100%	A/E/F	=	0.0	0.0	0.0	0.324	0.315

Section 4 — dt Sensitivity: Complete Data

Key finding: F fixes E's dpcc-c-tight-dt0.25 failure on both-hard (E=0%, F=100%). All other dt settings match between E and F.

Top-Right-Hard

dt	A G&C;	E G&C;	F G&C;	ΔEF	A Steps±std	F Steps±std	A Time	E Time	F Time
dt=0.25	100%	50%	50%	=	81.0±6.0	90.5±12.5	2.225	0.453	0.452
dt=0.5	0%	100%	100%	=	75.5±2.5	72.0±11.0	1.668	0.274	0.275
dt=1.0 (def)	100%	100%	100%	=	66.5±0.5	63.0±7.0	0.911	0.242	0.243
dt=2.0	100%	100%	100%	=	66.5±0.5	72.5±0.5	0.689	0.247	0.248
dt=4.0	50%	100%	100%	=	72.0±0.0	142.0±4.0	0.535	0.268	0.260

Top-Left-Hard

dt	A G&C;	E G&C;	F G&C;	ΔEF	A Steps±std	F Steps±std	A Time	E Time	F Time
dt=0.25	100%	50%	50%	=	70.0±3.0	137.5±37.5	1.259	2.627	2.577
dt=0.5	100%	100%	100%	=	63.5±0.5	76.0±9.0	0.937	0.517	0.523
dt=1.0 (def)	100%	100%	100%	=	63.5±3.5	67.0±7.0	0.822	0.347	0.344
dt=2.0	100%	100%	100%	=	64.5±0.5	69.5±3.5	0.679	0.259	0.267
dt=4.0	0%	100%	100%	=	107.0±2.0	142.5±8.5	0.810	0.246	0.248

Both-Hard

dt	A G&C;	E G&C;	F G&C;	ΔEF	A Steps±std	F Steps±std	A Time	E Time	F Time
dt=0.25	50%	0%	100%	+100 pp	75.5±0.5	71.5±3.5	1.835	0.693	0.572
dt=0.5	100%	100%	100%	=	68.5±4.5	60.5±5.5	1.185	0.585	0.556
dt=1.0 (def)	100%	100%	100%	=	56.5±2.5	61.5±4.5	0.997	0.405	0.401
dt=2.0	100%	100%	100%	=	64.5±2.5	72.0±14.0	0.737	0.336	0.319
dt=4.0	100%	100%	100%	=	85.0±2.0	96.0±20.0	0.691	0.324	0.315

Section 5 — Summary Statistics & ODE Step Analysis

5.1 Aggregate Statistics — Core Methods (13×3=39 observations each)

Metric	A (Diffusion)	E (ODE=1)	F (ODE=10)	F–E Δ	F–A Δ
G&C; Success	**0.551**	**0.449**	**0.487**	**+0.038**	**-0.064**
Constraint Sat.	**0.551**	**0.449**	**0.487**	**+0.038**	**-0.064**
Task Success	**0.987**	**0.846**	**0.923**	**+0.077**	**-0.064**
Avg Violations	**4.936**	**7.179**	**9.821**	**+2.641**	**+4.885**
Avg Total Viol.	**0.372**	**1.371**	**1.398**	**+0.027**	**+1.026**
Avg Time (s/step)	**0.671**	**0.274**	**0.261**	**-0.013**	**-0.409**

5.2 Win Counts — G&C; (39 comparisons)

A: 9	E: 0	F: 3	Ties: 27

5.3 Key Findings: ODE Steps 1 → 10

1. Increasing ODE steps has negligible effect on G&C; quality

E and F produce identical G&C; scores on 36/39 method×scenario pairs (92%). The FMv2 architecture already produces high-quality trajectories with a single ODE step. Only 2 differences exist: F fixes E's dpcc-c-tight-dt0.25 failure on both-hard (0%→100%), and F shows changed dpcc-r/dpcc-t behaviour on both-hard.

2. F resolves E's critical dt=0.25 failure on both-hard

E scores 0% G&C; at dpcc-c-tight-dt0.25 on both-hard, a notable regression. F recovers this to 100% (71.5 steps, clean). This is the single most meaningful improvement from ODE=10 — it removes an unstable edge case at very small dt.

3. F introduces a new regression: dpcc-r on both-hard worsens

E scores 0% G&C; on dpcc-r (both-hard), consistent with other FM variants. F scores 50% — superficially an improvement — but with 36.5 avg violations at 97.5 steps. Similarly dpcc-t on both-hard: E=0% (3.5 violations), F=0% (35.5 violations, 103 steps). F exhibits longer, more violation-prone trajectories on these methods.

4. ODE=10 adds negligible computation overhead

E averages 0.274s/step, F averages 0.261s/step — a difference of 12.6ms/step (4.6%). The FMv2 architecture's inference cost is dominated by the DPCC projection, not the ODE integration. Both E and F remain 2-3× faster than A (Diffusion) and B/C (original FM).

5. E and F are functionally equivalent for tightened DPCC methods

All tightened variants (dpcc-r-tight, dpcc-c-tight, dpcc-t-tight, post_proc-tight) score identically for E and F across all three scenarios. These remain the recommended methods for deployment.

6. Diffusion (A) retains quality advantages on baseline and untightened methods

A wins 9 comparisons vs F=3/E=0 (ties=27). A's diffuser/gradient/model_free methods achieve 100% G&C; on top-right-hard (0 violations), while E and F both produce 33-36 violations per episode on those methods.

7. Recommendation: E (ODE=1) is sufficient; F (ODE=10) offers no systematic advantage

The single meaningful improvement of F (fixing dt=0.25 on both-hard) does not justify the added complexity of 10 ODE steps when E already handles all other configurations correctly. For deployment: use FMv2 (E) with tightened DPCC methods (dpcc-c-tight, dpcc-t-tight) at dt=0.5 or dt=1.0 for the best balance of speed, quality, and constraint satisfaction.

A=Diffusion (diffusion/6). E=FMv2 (flow_matching_v2/6, aw=1, ODE steps=1, ckpt 90000). F=FMv2 (flow_matching_v2/6, aw=1, ODE steps=10, ckpt 90000). All Seed 6, avoiding-d3il, H8_K20_Dmodels.